



Hands-On Activity: Kaggle datasets



Activity overview

In the last activity, you got set up on Kaggle and explored the Notebooks feature. In this activity, we will work with a different feature of the Kaggle platform: datasets.

Kaggle has tens of thousands of datasets that are available for public use. Anyone can upload a dataset to Kaggle. If they choose to make it public, other Kagglers can use that dataset to create their own projects.

First, you'll take a tour of a specific dataset. Then, you'll have a chance to choose your own datasets to work with. Finally, you'll use what you've learned in this module to determine the kind of data in your datasets, and whether the data is biased or unbiased.

By the time you complete this activity, you will be able to use many of the helpful features Kaggle has to offer. This will enable you to find data for projects and engage with the data community, which is important for developing skills and networking in your career as a data analyst.

Explore Kaggle datasets

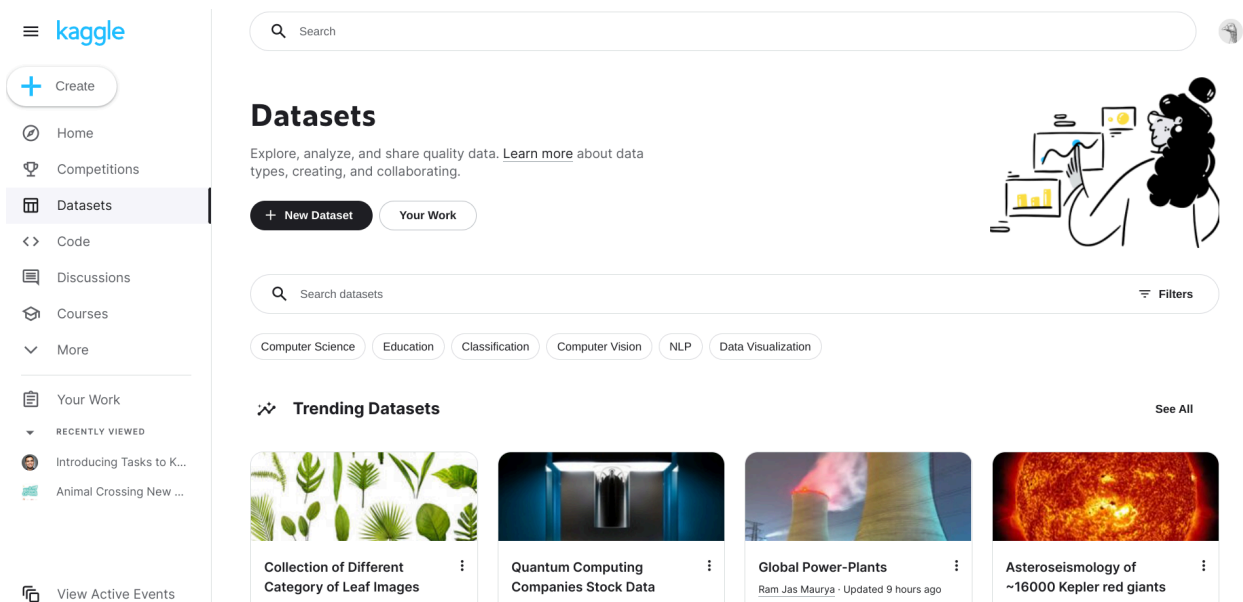
Let's explore the datasets feature!

Find a dataset

1. To start, log in to your Kaggle account.

- **Note:** Kaggle frequently updates its user interface. The latest changes may not be reflected in the screenshots, but the principles in this activity remain the same. Adapting to changes in software updates is an essential skill for data analysts, and we encourage you to practice troubleshooting. You can also reach out to your community of learners on the discussion forum for help.

2. Then, click on the **Datasets** icon in the vertical **Navigation** bar on the left. This takes you to the **Datasets** home page. From here, you can create a new dataset or search for datasets created by other Kagglers.



The screenshot shows the Kaggle Datasets homepage. On the left is a vertical navigation bar with the Kaggle logo and links to Home, Competitions, Datasets (highlighted), Code, Discussions, Courses, More, Your Work, and a Recently Viewed section. The main content area has a search bar at the top, followed by the 'Datasets' title and a brief description. Below this are buttons for 'New Dataset' and 'Your Work'. A secondary search bar and a 'Filters' button are also present. A row of category tags includes Computer Science, Education, Classification, Computer Vision, NLP, and Data Visualization. The 'Trending Datasets' section displays four featured datasets: 'Collection of Different Category of Leaf Images', 'Quantum Computing Companies Stock Data', 'Global Power-Plants' (updated 9 hours ago by Ram Jas Maurya), and 'Asteroeismology of ~16000 Kepler red giants'.

3. Now, check out a specific dataset. Enter **Animal Crossing** in the search bar to find datasets related to the Nintendo video game *Animal Crossing*.

4. There's more than one option, so click on the **Animal Crossing New Horizons Catalog**. This takes you to the landing page for this dataset.

Tour a dataset landing page

The screenshot shows the Kaggle website interface. On the left is a sidebar with navigation links: Home, Competitions, Datasets (highlighted), Code, Discussions, Courses, and More. Below these are 'Your Work' and 'RECENTLY VIEWED' items, including 'Introducing Tasks to K...' and 'Animal Crossing New ...'. The main content area features a search bar at the top. Below it is a banner for the 'Animal Crossing New Horizons Catalog' dataset by Jessica Li, updated 9 months ago (Version 3). The banner includes a 'Download (4 MB)' button and a 'New Notebook' button. Below the banner, there's a section for 'Usability 8.2', 'License CC0: Public Domain', and 'Tags simulations'. The 'Description' section is titled 'Context' and contains a paragraph about the dataset's origin and the game's details.

Header: The header at the top of the page contains the following information about the dataset:

- Its title
- A brief description of its contents
- The name of its creator
- When it was last updated
- Its current version

Badge: In the top-right corner of the header, you'll find three more items:

- A badge in the shape of a circle

- An icon in the shape of a caret symbol (^)
- A number

The badge is related to Kaggle's progression system. If you want, you can read more about it [here](#).

Upvotes: Clicking on the caret lets you “upvote” the dataset. The number shows the number of times this dataset has been upvoted by the Kaggle community.

Tabs: Beneath the header is a bar with three tabs: Data Card, Code, and Discussion. Take a moment to click on each of these tabs and explore their contents. Afterwards, navigate back to the Datasets tab!

Now, you can move down the page. You'll find a box that contains three terms:

Usability, License, and Tags.

Usability shows how complete the dataset *webpage* is (and not the dataset itself). Kaggle encourages the community to add information to the dataset webpage to make the dataset itself easier to understand. For example, a brief description or a column header. Hover your cursor over the **Usability score** to discover what the dataset page contains.

Licenses govern how a dataset can be used. Click on the **license name** to learn more about that specific license.

Expected update frequency will report when the data set was updated with new attributes, amendments, or expansions of the data. You may also see when any expected future dates are scheduled for a data set update. In this case, there will never be an update scheduled.

The next box down contains a detailed description of the dataset. Kagglers often include information on where the dataset came from and how the dataset was prepared.

And last—but certainly not least—is the Data Explorer!

Use the data explorer

The **Data Explorer** menu shows that the *Animal Crossing* dataset contains 30 .csv files. If you click on a file name, the window to the right will display information from that specific file. Try clicking on **umbrellas.csv** to check it out!

The screenshot shows the Kaggle Data Explorer interface. On the left is a sidebar with navigation options: Home, Competitions, Datasets (selected), Code, Discussions, Courses, and More. Below these are 'Your Work' and 'RECENTLY VIEWED' sections. The main area displays the 'Data Explorer' for the 'umbrellas.csv' file (9.4 kB). It includes a search bar, tabs for 'Data', 'Code (15)', 'Discussion (51)', 'Activity', and 'Metadata', and buttons for 'Download (4 MB)' and 'New Notebook'. The 'Data' tab is active, showing a 'Detail' view of the file. The 'About this file' section states 'All possible umbrellas to obtain'. Below this is a table with columns: Name, DIY, Buy, Sell, Color 1, and Color 2. The table lists various umbrella types like 'apple umbrella', 'bat umbrella', 'beach umbrella', 'bear umbrella', and 'black chic umbrella'. To the left of the table, there are summary statistics: 67 unique values for Name, a pie chart for DIY (true: 9, 13%; false: 58, 87%), a bar chart for Sell (80, 19%; 3500, 18%), and a bar chart for Color 1 (Green: 13%, Yellow: 12%, Other: 75%).

Tour the dataset explorer

Notice that the Data Explorer has three viewing options: **Detail**, **Compact**, and **Column**. For now, we'll focus on the **Detail** tab.

The description at the top of the **Detail** tab shows that the **umbrellas.csv** file contains data on all the umbrellas in the video game. Let's check out the columns. Each column header has three items:

- A small icon on the left that shows the data type
- The name of the column
- An icon with three bars that lets you sort the data if you click on it

Below each column header is a box that contains a summary of the data. This lets you quickly get an idea of what's in the dataset. For example, the summary for the **Name** column shows there are 67 unique values for the umbrella names. The summary for the **DIY** column shows that 9 of the umbrella recipes are DIY, or "do it yourself." Take a moment to explore the summaries for the other columns.

And that completes our tour! That's a lot of information. Feel free to go back and review.

Access a dataset

After you've explored a dataset, you can link it to a Kaggle notebook or download it to access it for your own use. Linking a dataset to a Kaggle notebook means you create a new notebook from the existing dataset so that it is available for you to use.

Find your own datasets

Now, you'll get a chance to choose your own datasets to work with! Use the following steps to find datasets that interest you:

1. When you're ready, click on the **Data** icon on the left to return to the **Datasets** landing page.
2. Note that datasets can exist in a variety of formats. If you want to make sure your dataset is in a .csv format, click on the **Filter** button on the right side of the Datasets search bar. Then, choose **CSV** from the menu.
3. Find 2-3 datasets that you're interested in exploring further.
4. Create notebooks from them, download them, or check them out in the Data Explorer. Keep these datasets in mind for your upcoming reflection.

Link or download a dataset



Here are the options to create a notebook or download the dataset:

- Create a Kaggle notebook: To link a dataset to a Kaggle notebook, you click on the **New Notebook** button in the dataset header. This will create a notebook in your Kaggle account that links to the dataset.
- Download the dataset: To download a copy of the dataset to your computer, click on the **Download** button in the dataset header at the top of the page.
- Open the file in Google Sheets: To open a Google Sheets view of the file, click on the **download** icon at the top-right of the Data Explorer. You can then download the file.

Reflection

Kaggle's datasets and Data Explorer allow you to do which tasks?

- ☐ Create visualizations from datasets
- ☒ Search for datasets

Correct

Kaggle's datasets and Data Explorer allow you to search for, access, and upload your own datasets. You can use Kaggle to conduct research, complete data projects, and share your accomplishments with other members of the data science community.

- ☒ Access datasets

Correct

Kaggle's datasets and Data Explorer allow you to search for, access, and upload your own datasets. You can use Kaggle to conduct research, complete data projects, and share your accomplishments with other members of the data science community.

- ☒ Upload your own datasets

Correct

Kaggle's datasets and Data Explorer allow you to search for, access, and upload your own datasets. You can use Kaggle to conduct research, complete data projects, and share your accomplishments with other members of the data science community.

2.

Question 2

So far, you've learned a lot about how to use Kaggle to explore datasets. During this activity, you used this knowledge to find datasets you're interested in. Keep those datasets in mind and in the text box below, write 2-3 sentences (40-60 words) in response to each of the following questions:

- What type(s) of data are in this dataset? Is this dataset biased or unbiased? How do you know?
- Based on what you've explored so far, how might you use Kaggle's Datasets feature to help develop your data analysis skills?

Answer:

There is structured data in the dataset. All the data are unbiased. Bias has evolved to become preference in favor of or against a person, group of people, or thing. But in here, no data of this type are found; all data are good data with a properly maintained structure. Kaggle is a big platform for datasets, and you can also develop and publish your own dataset. You can also explore someone's dataset and edit it for new research purposes and publish it as a completely new type of dataset, which is totally unique. Kaggle is a very useful site for data enthusiasts.

Feedback

Congratulations on completing this hands-on activity! A strong response would include that Kaggle gives you access to tons of public datasets that you can use to practice data analysis and create your own projects. Beyond that, consider the following:

Online platforms like Kaggle allow you to search for, view, explore, upload, and work with datasets from a variety of sources and perspectives. Understanding how Kaggle works and how to use it will help you develop your skillset and grow as a data analyst.